

Embedding vs referencing

Embedding stores related data within a parent document as nested subdocuments/arrays. This keeps data physically together, and is useful when either when operating with related information that is accessed together or when dealing with exclusive data.

On the other hand, referencing involves storing identifiers/references that point to documents in other collections, making the related data remain physically separated but still logically connected. It is often used when handling shared or frequently changed/updated data.

APPLICATION TO OUR MUSIC PLATFORM DATABASE

USERS

- Main strategy: embedding
- Embedded field: subscription
- Justification: subscription data represents a one-to-one relationship in which each user has its unique/exclusive subscription plan and details . This information is consistently accessed through the user's profile and has a low update frequency. Embedding eliminates the need for additional queries when retrieving user profiles with subscription status.

PLAYLIST

- Main strategy: referencing
- Referenced fields: user_id, songs
- Justification:
 - user_id: references the creator of the playlist in the 'users' collection. This way, we prevent data duplication and ensure consistency in the sense that updates or changes in a user's profile propagate automatically to all playlists created by the user, without having to manually change information in each of them.
 - song_ids: instead of embedding song details, we considered creating references to songs. Since songs are shared entities, meaning they could appear in multiple playlists, referencing allows centralized management of the song's data, which implies that the song's information could be updated one single time.

PLAY_HISTORY

- Main strategy: referencing
- Referenced fields: user_id, song_id, playlist_id
- Justification: active users can generate thousands of playback records. Embedding all of this data within play_history could potentially risk exceeding MongoDB's document size limit. Instead, by using references, we can have access to complete information when required, while still maintaining the document size within limits.

FOLLOWS

- Main strategy: referencing
- Referenced fields: followers, following_users
- Justification: social relationships in any social platform represent a many-to-many relationship, where users can follow and be followed by multiple others. Thus, referencing enables efficient queries in both directions -finding a user's followers and who they follow- while preventing data duplication that would result of embedding.